



Physiologically Based Pharmacokinetic Modeling of Vedolizumab in Healthy Adults and Patients with Inflammatory Bowel Disease

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Abstract

Background and Objective Vedolizumab is a gut-selective, humanized monoclonal antibody that binds to $\alpha 4\beta 7$ integrin, preventing lymphocyte trafficking to the intestinal mucosa. While approved for adults with moderate-to-severe ulcerative colitis (UC) or Crohn's disease (CD), drug exposure may vary significantly with disease activity. Physiologically based pharmacokinetic (PBPK) modeling provides a mechanistic framework to integrate drug properties with system physiology, enabling the prediction of pharmacokinetics (PK) under these various pathophysiological conditions. In this study, we developed a minimal mechanistic PBPK model incorporating target-mediated drug disposition (TMDD) to characterize the PK of healthy adults and patients with varying inflammatory bowel disease severity.

Methods A minimal PBPK model incorporating TMDD was developed for vedolizumab using the Simcyp Simulator. The model was parameterized with literature values for healthy adults, verified against clinical PK data from healthy volunteers, and adapted for patients with varying disease severity by estimating target abundance ($R_{\max}$) to reflect $\alpha 4\beta 7$ upregulation in active inflammation. Model performance was assessed by comparing simulated and observed concentration–time profiles and predicted-to-observed (P/O) ratios for peak concentration ($C_{\max}$), area under the curve from time 0 to time of the last concentration ($AUC_{0-t_{\text{last}}}$), and trough concentration (C_{trough}).

Results The PBPK model demonstrated robust predictive performance across varying disease states and ethnicities, compared with reported PK parameters in clinical studies. In healthy adults, mean P/O ratios for $C_{\max}$ and $AUC_{0-t_{\text{last}}}$ ranged from 0.81 to 1.11 and 0.80 to 1.02, respectively, across single doses of 0.2–10 mg/kg. For patients with mild-to-moderate UC, the estimated $R_{\max}$ yielded mean P/O ratios for $C_{\max}$ and $AUC_{0-t_{\text{last}}}$ of 0.88–1.02 and 0.78–0.99, respectively. Additionally, for moderate-to-severe UC and CD, mean P/O ratios for C_{trough} fell within 0.67–1.49. Simulations indicated that higher $R_{\max}$ values increased clearance and reduced systemic exposure, with mean predicted C_{trough} in high target-expression groups frequently falling below the efficacy thresholds of 12 $\mu\text{g/mL}$ and 14 $\mu\text{g/mL}$.

Conclusions This study presents the first PBPK model for vedolizumab that quantitatively captures disease-related $\alpha 4\beta 7$ upregulation and its impact on PK in adults. The model provides a mechanistic basis for simulating exposure across disease states and supports model-informed precision dosing strategies to optimize treatment outcomes in UC and CD.

1 Introduction

Inflammatory bowel disease (IBD), comprising ulcerative colitis (UC) and Crohn's disease (CD), is a chronic, relapsing inflammatory disorder of the gastrointestinal tract

that can cause progressive intestinal injury, impair quality of life, and impose a substantial healthcare burden [1, 2]. Effective long-term management is crucial for inducing and maintaining remission, preventing complications, and enhancing patient outcomes [1, 3, 4]. In moderate to severe

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Key Points

This study developed a physiologically based pharmacokinetic (PBPK) model incorporating target-mediated drug disposition (TMDD) to characterize vedolizumab pharmacokinetics (PK) across different inflammatory states in inflammatory bowel disease.

By estimating disease-dependent $\alpha 4\beta 7$ target abundance, the model successfully reproduced observed PK profiles and demonstrated that increased target expression can enhance clearance and reduce systemic exposure.

These findings provide a mechanistic framework to support model-informed precision dosing of vedolizumab.

disease, biologic therapies have revolutionized treatment by enabling the targeted modulation of immune pathways that drive intestinal inflammation [1, 5].

Vedolizumab is a gut-selective, humanized monoclonal antibody that binds to the $\alpha 4\beta 7$ integrin, thereby blocking lymphocyte trafficking to the intestinal mucosa. The integrin $\alpha 4\beta 7$, expressed on T cells, interacts with the cell-adhesion molecule MAdCAM-1 present on the endothelial cells of the gut. Vedolizumab disrupts this interaction, preventing lymphocyte infiltration into tissues [6–8]. This tissue-specific mechanism distinguishes vedolizumab from systemic immunosuppressants and contributes to a favorable safety profile [9–11]. It is an established treatment option for adults with moderate to severe UC or CD who have experienced inadequate response or intolerance to conventional therapies or anti-tumor necrosis factor (anti-TNF) agents. Clinical trials have demonstrated efficacy in both induction and maintenance of remission, and fixed-dose regimens are widely implemented in adult practice [12–14].

For other monoclonal antibodies used in IBD, such as infliximab and adalimumab, numerous studies have demonstrated a relationship between serum drug concentrations and therapeutic outcomes [15, 16]. Therapeutic drug monitoring (TDM) is frequently employed to guide individualized dosing for these agents, as drug exposure is often influenced by patient- and disease-related factors such as inflammatory burden, protein loss, and target expression [17, 18]. Emerging evidence suggests that vedolizumab exposure is similarly affected by disease activity and is associated with treatment response [19, 20].

Physiologically based pharmacokinetic (PBPK) modeling provides a mechanistic framework to address these variabilities. By integrating drug-specific properties with system-level physiological information, PBPK models enable the simulation of pharmacokinetics (PK) under varying

pathophysiological conditions [21]. As such, PBPK models may complement TDM by supporting the prediction of drug exposure and informing model-informed precision dosing (MIPD) strategies. For infliximab, the PBPK modeling framework has successfully quantified the impact of disease-related alterations, including increased protein catabolism, changes in vascular permeability, and modulation of target expression on PK [22]. Such models support dose optimization and inform dosing strategies in subpopulations with limited clinical data, including pediatric patients [23, 24]. Despite vedolizumab's established role in IBD management, no PBPK model currently exists to mechanistically evaluate the influence of disease severity on its disposition. The aim of this study was to address this gap by developing and qualifying a PBPK model for vedolizumab in healthy adults and adapting it for patients by incorporating physiological alterations associated with varying IBD severity. This work establishes a foundation for extending model-informed dosing strategies to other subpopulations, including pediatric patients, where direct PK data remain scarce.

2 Methods

2.1 PBPK Model Structure

A minimal mechanistic PBPK model for vedolizumab was implemented in the Simcyp Simulator (version 22.0, Certara, Sheffield, UK), using the platform's population and substrate building blocks for model modification and prediction. The minimal PBPK (mPBPK) model was selected to characterize vedolizumab PK while reducing the risk of over-parameterization by incorporating only the key physiological processes governing antibody disposition. The model consists of blood, tissue, and lymph node compartments. The blood compartment is subdivided into plasma and blood cells sub-compartments and parameterized using physiological characteristics such as plasma volume, hematocrit, and cardiac output. The tissue compartment is further divided into vascular, endosomal, interstitial, and intracellular spaces. A schematic representation of the mPBPK framework is shown in Fig. 1. The mPBPK framework incorporates several key processes governing antibody disposition: (1) drug distribution from the arterial blood into the tissue vascular space and return through venous blood and lymphatic flow; (2) convective transport between vascular and interstitial spaces mediated by endothelial pores and vascular reflection coefficients; (3) endosomal uptake and neonatal Fc receptor (FcRn)-mediated recycling of the drug and endogenous immunoglobulin G (IgG); (4) lysosomal degradation of unbound drug within the endosomal compartment; and (5) target-mediated drug disposition (TMDD), including drug–target

binding, internalization, and clearance of the drug–target complex. Within the Simcyp Simulator, TMDD processes can be implemented in plasma, tissue, or lymph compartments depending on the target distribution assumptions. In TMDD, the dynamic target binding capacity refers to the time-varying changes in target abundance ($R_{\max}$), driven by the interplay between synthesis and degradation rates of the target molecule. The mathematical equations describing the mPBPK model framework, with the quasi-equilibrium (QE)-TMDD approximation in the Simcyp simulator, are mentioned in detail in previous publications [25–27].

2.2 Clinical Data Sources

Clinical PK data were obtained from published phase I–III studies in healthy adults and adult patients with UC or CD. Data included single-dose escalation studies in healthy volunteers and multiple-dose studies stratified by disease severity [28–33]. Further model validation was performed using PK data of another ethnic population (Japanese patients) [28, 34–37]. Published concentration–time profiles were digitized using WebPlotDigitizer (Version 4.8; <https://apps.automeris.io/wpd4/>) to extract numerical values for model development and evaluation.

2.3 Model Development in Healthy Adults

An mPBPK model with a 1:1 FcRn binding stoichiometry within the endosomal compartment was used to predict the vedolizumab concentration–time profile. Within the mPBPK model framework (Fig. 1), the parameter values defined for endogenous IgG parameters (e.g., vascular reflection coefficients [σ_{vs} , σ_{vL}], rate of uptake into endosomal space [K_{up}], recycling rate of FcRn-bound vedolizumab [K_{rc}], catabolic clearance [CL_{cat}], and additional systemic clearance [CL_{add}]) were used as the baseline parameter values to initialize the vedolizumab model [25]. The value of the vedolizumab-specific parameter, such as the equilibrium binding rate constant (K_D), which defines the drug’s affinity for FcRn, was obtained from in vitro studies [38]. Additionally, the dynamic QE-TMDD model was considered, using parameters including $\alpha 4\beta 7$ $R_{\max}$, degradation rate constant (K_{deg}), binding dissociation constant (K_D), and drug–target complex internalization rate constant (K_{int}). These parameter values were obtained from the literature on antibodies with similar pharmacological and target characteristics [39]. All parameters, with their corresponding values and sources, are detailed in Table 1. Distribution and clearance parameter values were refined by fitting simulations to healthy

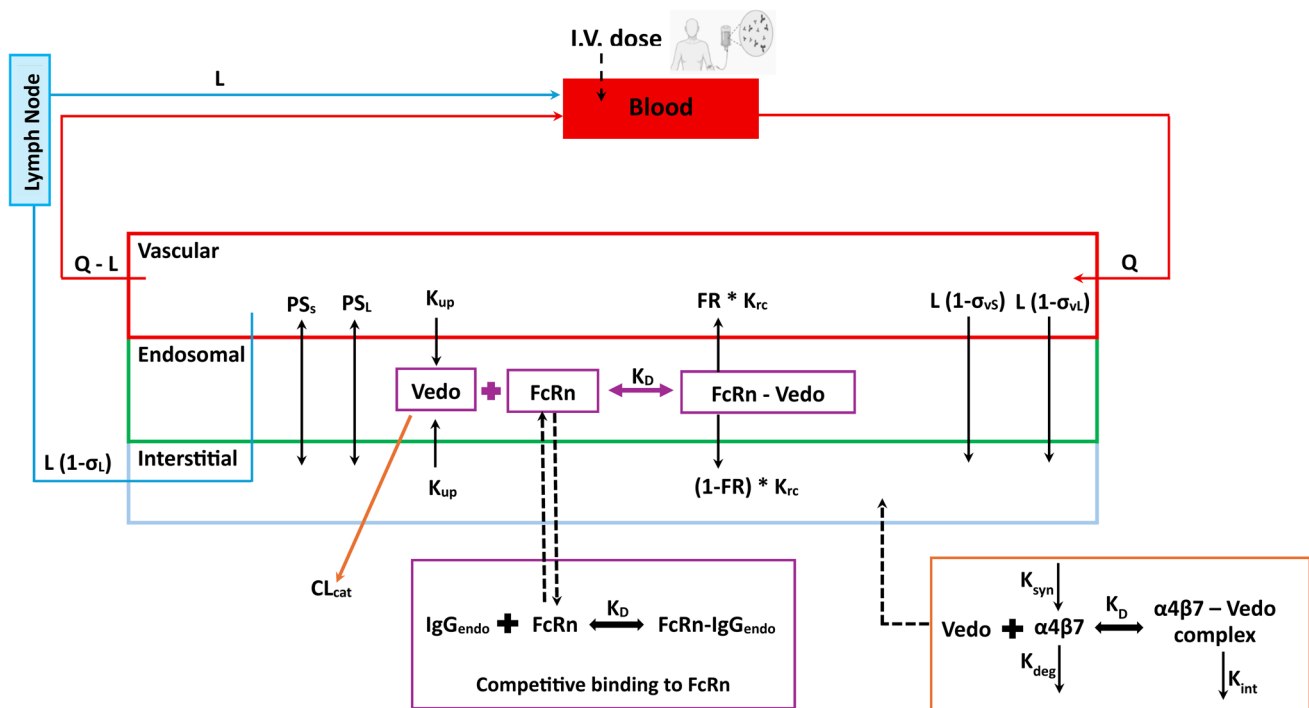


Fig. 1 A minimal PBPK model structure illustrating TMDD and the key mechanisms for vedolizumab disposition (adapted from Simcyp)^{25,47}. CL_{cat} an unbound IgG lysosome-mediated catabolic clearance, $FcRn$ neonatal Fc receptor, K_D equilibrium binding rate constant of the drug with FcRn, FR recycling fraction, IgG immunoglobulin G, K_{deg} target degradation rate constant, K_D equilibrium dis-

sociation constant, K_{int} rate constant for drug–target complex internalization, K_{rc} FcRn-bound IgG-recycling rate, K_{syn} rate of synthesis of target, K_{up} fluid-phase endocytosis rate, $PBPK$ physiologically based pharmacokinetic, $TMDD$ target-mediated drug disposition, σ_{vL} vascular reflection coefficient of large pores, σ_{vs} vascular reflection coefficient of small pores

volunteer PK data using the Simcyp Healthy Adult population file, employing the weighted least squares method for parameter estimation.

2.4 Model Development in Adult Patients with UC and CD

The verified healthy adult model was adapted for UC and CD populations by incorporating disease-related physiological changes, specifically the upregulation of $\alpha 4\beta 7^+CD4^+$ T cells.

The effective R_{max} was estimated to reflect differences in target levels across disease severity categories. Disease severity was stratified based on the clinical scoring indices: the Mayo endoscopic sub-score for UC, and the Crohn's Disease Activity Index (CDAI) for CD. To match the demographics of the clinical trial cohorts, the “Sim-Healthy Volunteers” and “Sim-Japanese” population libraries provided in the Simcyp Simulator were used and optimized for patients with target-concentration changes.

Table 1 Input parameter values for the vedolizumab PBPK model

Parameter	Value	%CV ^b	Source
Lymph flow rate (L/h/kg)	0.0017	–	[25, 40]
Molecular weight (kDa)	146.8	–	[8]
Hydrodynamic radius (nm)	5.04 ^a	–	Calculated
Fraction unbound in plasma	1 ^b	–	Simcyp default
B:P	0.55 ^b	–	Simcyp default
Human serum albumin (g/L)	45 ^b	–	Simcyp default
K_D (μ M)	0.61	10	[38]
K_{up} (1/h)	0.00302 ^c	–	Estimated
K_{rc} (1/h)	3.13	–	Assumed same as endogenous IgG
FR	0.5	–	Assumed same as endogenous IgG
σ_{vs}	0.932 ^c	–	Estimated
σ_{vL}	0.172 ^c	–	Estimated
CL_{cat} (L/h)	0.0175	10	Assumed same as endogenous IgG
CL_{add} (L/h)	0.0052 ^d	30	Estimated
<i>Quasi-equilibrium target-mediated drug disposition model</i>			
R_{max} (μ M)			Estimated
Healthy	0.0027 ^e	30	
Mild-to-moderate	0.0048 ^e	30	
Moderate-to-severe	0.0075 ^f	30	
K_{deg} (1/h)	0.0061 ^e	30	Estimated
K_{syn} (μ M/h)			–
Healthy	1.7E–05 ^g	30	
Mild-to-Moderate	2.9E–05 ^g	30	
Moderate-to-Severe	4.6E–05 ^g	30	
K_D (μ M)	1.3E–05	20	[39]
K_{int} (1/h)	0.000896	20	[39]

B:P blood to plasma ratio, CL_{add} additional systemic clearance, CL_{cat} an unbound IgG lysosome-mediated catabolic clearance, *CV* coefficient of variation, *FcRn* neonatal Fc receptor, *FR* recycling fraction, *IgG* immunoglobulin G, K_D equilibrium binding rate constant of drug with FcRn, K_D equilibrium dissociation constant, K_{deg} target degradation rate constant, K_{int} rate constant for drug–target complex internalization, K_{rc} FcRn-bound drug-recycling rate, K_{syn} rate of synthesis of target, K_{up} fluid-phase endocytosis rate, *PBPK* physiologically-based pharmacokinetic, *PK* pharmacokinetic, R_{max} target abundance, σ_{vs} vascular reflection coefficient of large pores, σ_{vL} vascular reflection coefficient of small pores

^aCalculated based on the molecular weight (g/mol) of vedolizumab in Simcyp[8]

^bSimcyp default values

^cEstimated by parameter estimation using PK data of 0.2-mg/kg dose from Rosario et al. [29]

^dEstimated by parameter estimation using PK data of 10-mg/kg dose from Rosario et al. [29]

^eEstimated by parameter estimation using PK data of 2-mg/kg dose from Parikh et al. [31]

^fEstimated by parameter estimation using PK data of a 300-mg dose from Feagan et al. [32]

^gCalculated based on R_{max} and K_{deg}

2.5 PBPK Model Verification

The developed model was verified against observed vedolizumab PK data from both healthy and diseased populations. Virtual trial designs were matched to the demographics (age range and female proportions) of corresponding clinical studies [28–37]. Model performance was evaluated using ten repeated trials with ten virtual individuals each ($n = 100$ per scenario) to characterize population variability and generate stable PK prediction intervals. This simulation strategy was selected to ensure numerical stability of the simulated PK profiles rather than to reproduce the exact sample size of the external evaluation datasets. The mean simulated profiles with the 2.5th–97.5th percentile prediction intervals were compared against the observed data digitized from the mean PK profile of the published clinical trials. Predictive performance was assessed using predicted-to-observed (P/O) ratios for the area under the curve from time 0 to time of the last concentration ($AUC_{0-tlast}$), peak concentration (C_{max}), and trough concentration (C_{trough}). Model acceptability was defined as a P/O ratio falling within the 0.5- to 2.0-fold range. To assess generalizability across ethnicities, external validation was performed using the verified adult patient model applied to the Simcyp Japanese population file.

2.6 Impact of R_{max} on Vedolizumab PK

The verified adult PBPK models with TMDD were used to simulate the impact of R_{max} on vedolizumab PK under a standard clinical maintenance regimen (intravenous [IV] 300 mg every 8 weeks). Concentration–time profiles were predicted for adult populations with varying disease severity to quantify C_{trough} across a range of estimated R_{max} values.

3 Results

3.1 Healthy Adult Model

The developed PBPK model was verified using clinical data obtained from healthy adults following single IV doses of vedolizumab, ranging from 0.2 to 10 mg/kg. The initial model, in which substrate distribution and clearance parameter values were defined using endogenous IgG values, underpredicted vedolizumab exposure (data not shown). To improve alignment with clinical observations, the lymph flow rate [25, 40], K_{up} , R_{max} , K_{deg} , σ_{vL} , σ_{vs} , and CL_{add} were sequentially optimized. Final parameter estimates are presented in Table 1, with R_{max} estimated at 0.0027 μ M in healthy adults based on observed clinical data. Model performance was verified across the 0.2–10 mg/kg dose range, where mean concentration–time profiles (with 2.5th and 97.5th percentile curves) accurately

captured the observed clinical data (Fig. 2). The mean P/O ratios for $AUC_{0-tlast}$ and C_{max} were 0.80–1.02 and 0.81–1.11, respectively (Table 2), all of which fell consistently within a 0.5- to 2-fold range. These results demonstrate the robust predictive performance of the mPBPK model and highlight the importance of TMDD to accurately represent vedolizumab disposition in healthy adults.

3.2 Adult Patient Model

When the verified healthy adult model was applied to patient PK data, vedolizumab concentration–time profiles and P/O ratios were slightly overestimated. Minor deviations observed in the terminal phase of certain patient groups likely reflected variability in target-mediated clearance or disease-related physiological changes not initially accounted for. To address these differences, R_{max} was re-estimated across disease severity groups. As detailed in Table 1, the estimated R_{max} values were 0.0048 μ M for patients with mild-to-moderate UC and 0.0075 μ M for those with moderate-to-severe UC or CD, compared with 0.0027 μ M for healthy volunteers. These values showing upregulation of $\alpha_4\beta_7$ expression in gut tissue during inflammation are consistent with previous studies that reported elevated target levels in patients compared with controls [41–43]. Incorporating these estimates improved the model fit (Figs. 3 and 4) and yielded P/O ratios consistent with clinical data across dosing regimens and populations (Tables 3 and 4). The model was further evaluated for generalizability using an external dataset representing a diverse ethnic population (Japanese patients) (Supplementary Tables S1 and S2; see the electronic supplementary material). The model demonstrated good robustness, with simulated mean PK profiles closely aligning with reported mean observed data (Supplementary Figs. S1 and S2). These results support the model's applicability in the Japanese population across varying disease severity levels. Overall, the PBPK models demonstrated acceptable predictive performance and were considered fit for purpose for subsequent simulation and sensitivity analyses.

3.3 Impact of R_{max}

Using the estimated R_{max} values, concentration–time profiles were simulated for the approved adult dose of 300 mg across three population groups: healthy adults, mild-to-moderate UC, and moderate-to-severe UC or CD. While the simulated $AUC_{0-tlast}$ and C_{trough} largely overlapped among groups, lower C_{trough} levels were observed, consistent with the corresponding lower exposure in the moderate-to-severe group (Fig. 5). Reported exposure–response analyses have suggested that maintenance vedolizumab C_{trough} of $\geq 9 \mu$ g/

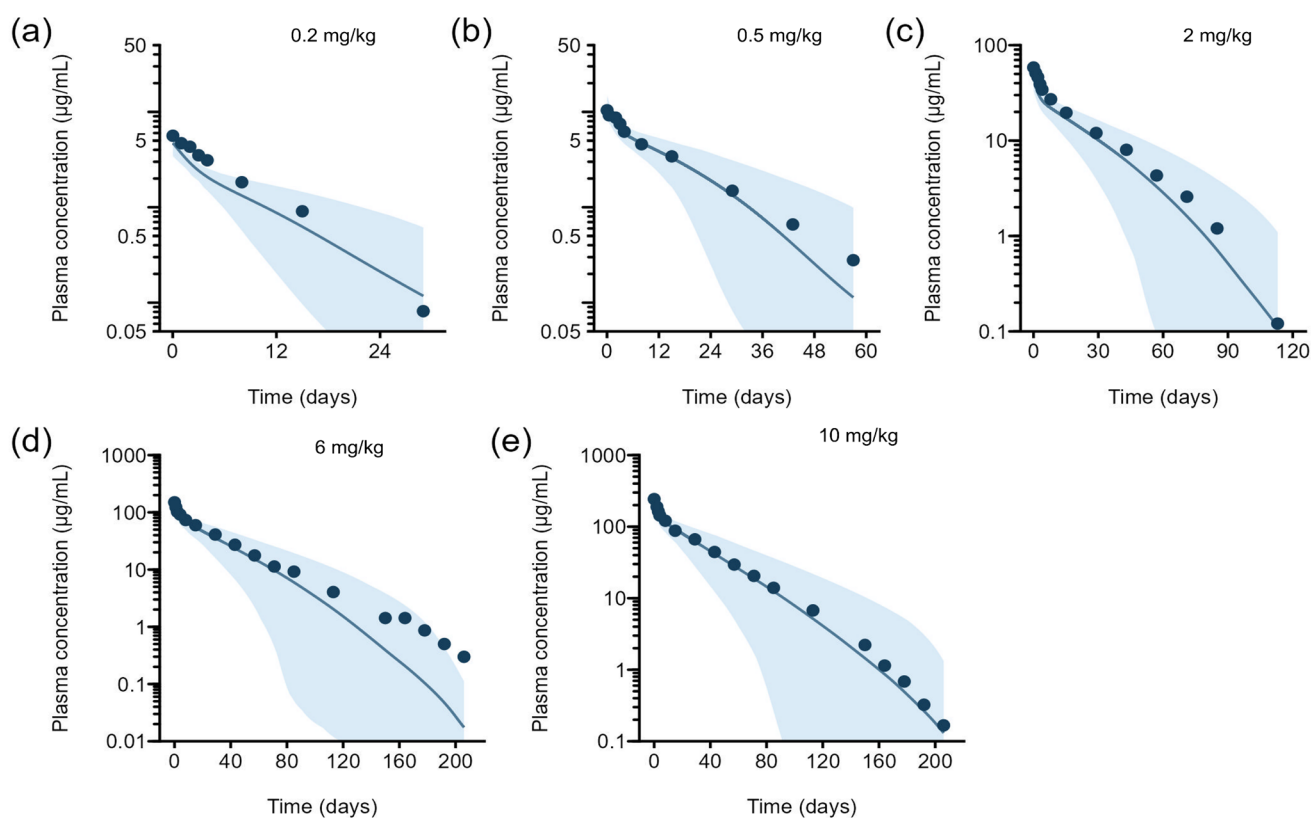


Fig. 2 Observed vs. predicted vedolizumab concentrations after a single IV dose in healthy adults. *Closed circles* represent the mean observed data²⁹. *Solid lines and shaded areas* indicate the predicted mean and 95% prediction interval ($n = 100$ per dose group). *IV* intravenous

mL and ≥ 12 $\mu\text{g/mL}$ are associated with clinical remission, while levels > 14 $\mu\text{g/mL}$ may be associated with improved clinical and endoscopic outcomes. These therapeutic target trough levels for vedolizumab were established using post hoc analyses of GEMINI clinical trials in patients with moderate-to-severe IBD [44, 45]. The simulated C_{trough} levels ($n = 100$ per group) across groups indicate that both patient population groups achieved C_{trough} levels below the therapeutic target threshold, with the lowest levels observed in severe patients (Fig. 5). These findings suggest that the MIPD approach, similar to that used for infliximab, may be beneficial for individualizing the dose to maintain C_{trough} within the target in patients with high disease burden.

4 Discussion

In this study, we developed an mPBPK model to describe vedolizumab PK in healthy adults and adult patients with IBD. The model was verified using clinical data from healthy individuals, patients with mild-to-moderate UC, and patients with moderate-to-severe UC or CD [28–37]. By incorporating disease-related upregulation of $\alpha 4\beta 7$ expression as a physiological factor, the model predicted PK profiles across

different populations. Simulations of a 300-mg IV dose administered every 8 weeks revealed that disease severity impacted both $\text{AUC}_{0-\text{tlast}}$ and C_{trough} .

PBPK modeling of large molecules, including monoclonal antibodies such as infliximab, ustekinumab, atezolizumab, and trastuzumab, has been used to quantify the impact of TMDD, disease-related physiological changes, and patient-specific variability on PK [22, 46–48]. However, this study presents the first PBPK model specifically developed for vedolizumab. By mechanistically capturing $\alpha 4\beta 7$ upregulation, the model elucidates the driver behind reduced exposure in patients with more severe disease and enables prediction of PK across disease states. Furthermore, with the inclusion of target-mediated physiological processes, the simulated total clearance at maintenance dosing ($\text{CL} = \text{dose}/\text{AUC}_{0-\text{tlast}}$) demonstrated an inter-individual variability of 32% (% coefficient of variation [%CV]) among virtual patients with moderate to severe disease. This estimate is consistent with previously reported clinical findings. A population PK analysis in adults using a two-compartment model reported inter-individual variability of 34.6% (%CV) in the linear clearance component of vedolizumab [49]. Similarly, a pediatric population PK study estimated clearance

Table 2 Simulated and observed vedolizumab PK parameters after a single IV dose in healthy adults

Dose (mg/kg)		$C_{\max 1d}$ ($\mu\text{g/mL}$)	$\text{AUC}_{0-\text{last}}$ ($\mu\text{g}\cdot\text{day/mL}$)
0.2	Observed ^a	5.62	31.3
	Predicted	4.70	28.9
	P/O ratio^b	0.84	0.92
0.5	Observed ^a	10.4	119
	Predicted	11.5	121.7
	P/O ratio^b	1.11	1.02
2.0	Observed ^a	58.4	955
	Predicted	47.6	762.2
	P/O ratio^b	0.81	0.80
6.0	Observed ^a	150	3020
	Predicted	141.8	2714.1
	P/O ratio^b	0.94	0.90
10.0	Observed ^a	243	4840
	Predicted	235.6	4804.9
	P/O ratio^b	0.97	0.99

$\text{AUC}_{0-\text{last}}$ area under the curve from time 0 to time of the last concentration, $C_{\max 1d}$ peak concentration at day 1, IV intravenous, PK pharmacokinetic, P/O predicted-to-observed

^aData from Rosario et al. [29]

^bP/O ratio calculated from the means of observed and simulated PK parameters

variability of 21.9% (%CV), incorporating body weight, serum albumin, and erythrocyte sedimentation rate (ESR) as covariates [50]. The concordance between PBPK-simulated variability and observed clinical data supports the physiological plausibility and predictive performance of the model. Taken together, this mechanistic PBPK framework provides a foundation for extrapolation to special populations with limited clinical data, including pediatric patients [21]. PBPK modeling has emerged as a valuable approach for MIPD, as it integrates physiological, physicochemical, and biological determinants to support mechanistically informed PK predictions beyond empirical modeling approaches. This framework is particularly useful for optimizing dosing strategies in special populations and informing initial dose selection. In addition, PBPK models can predict tissue target-site concentrations by accounting for physiological determinants of drug distribution. Despite these advantages, the clinical translation of PBPK modeling remains limited, and further studies are needed to better define its practical utility and validate it in real-world MIPD settings [51].

The estimated differences in $R_{\max}$ corroborate the biological understanding that $\alpha 4\beta 7$ expression varies with disease activity. $\alpha 4\beta 7$ is a gut-homing integrin expressed on activated lymphocytes, promoting migration to the intestinal

mucosa via binding to mucosal addressin cell adhesion molecule-1 (MAdCAM-1). Inflammatory activity in UC and CD is associated with increased $\alpha 4\beta 7$ expression on circulating T cells, reflecting lymphocyte activation and recruitment to inflamed tissue [41–43]. These model estimates are consistent with a previous report from Meenan et al., which showed elevated levels of activated gut-homing T cells (CD4+CD25+ $\alpha 4\beta 7$ +) in IBD patients [41]. By quantitatively representing this mechanism, our model effectively links disease severity to altered PK via TMDD.

Simulation results indicate that higher $R_{\max}$ increases target–drug binding and accelerates clearance, thereby reducing systemic drug exposure. While overall exposure metrics such as $\text{AUC}_{0-\text{last}}$ are commonly used in PK analyses to evaluate systemic drug exposure, they are not typically used for routine dose individualization in clinical practice. In contrast, C_{trough} is more directly applicable to clinical TDM and is the PK parameter most evaluated in studies of vedolizumab in patients with IBD. Exposure–response analyses have suggested that maintenance vedolizumab C_{trough} levels are associated with favorable clinical outcomes, although the proposed target thresholds vary across studies. A systematic review and meta-analysis of five cohort studies identified target vedolizumab C_{trough} of ≥ 9.0 – $12.6 \mu\text{g/mL}$ during maintenance as associated with improved clinical outcomes, with concentrations $\geq 12.6 \mu\text{g/mL}$ potentially conferring additional benefit [44]. A post hoc analysis of clinical trial data proposed a maintenance target of $12.7 \mu\text{g/mL}$ in patients with UC, whereas real-world cohort studies have reported thresholds exceeding $14 \mu\text{g/mL}$ during maintenance therapy [19, 45]. In contrast, another real-world retrospective study suggested a broader therapeutic range of 5 – $15 \mu\text{g/mL}$ in clinical practice [52]. Given this variability across clinical trials and real-world settings, the present study adopts a pragmatic reference range, with a mean maintenance C_{trough} of $\geq 9 \mu\text{g/mL}$ and $\geq 12 \mu\text{g/mL}$ considered indicative of adequate therapeutic response, and a maintenance target C_{trough} of $> 14 \mu\text{g/mL}$ potentially associated with improved clinical outcomes. Our simulations predict that patients with high target expression (i.e., severe disease) exhibit lower exposures, potentially increasing the risk of falling below these proposed efficacy thresholds. These findings support the potential value of model-informed or stratified dosing strategies to maintain adequate drug exposure across patient populations.

This study has several limitations. Parameter non-identifiability remains a structural challenge in PBPK modeling, particularly for mechanisms such as antibody lymphatic transport, distribution, and recycling when only plasma PK data are available. Although parameter non-identifiability cannot be fully resolved, we addressed this limitation in this study by fixing certain parameters based on published data. Sensitivity analysis was then used to identify which

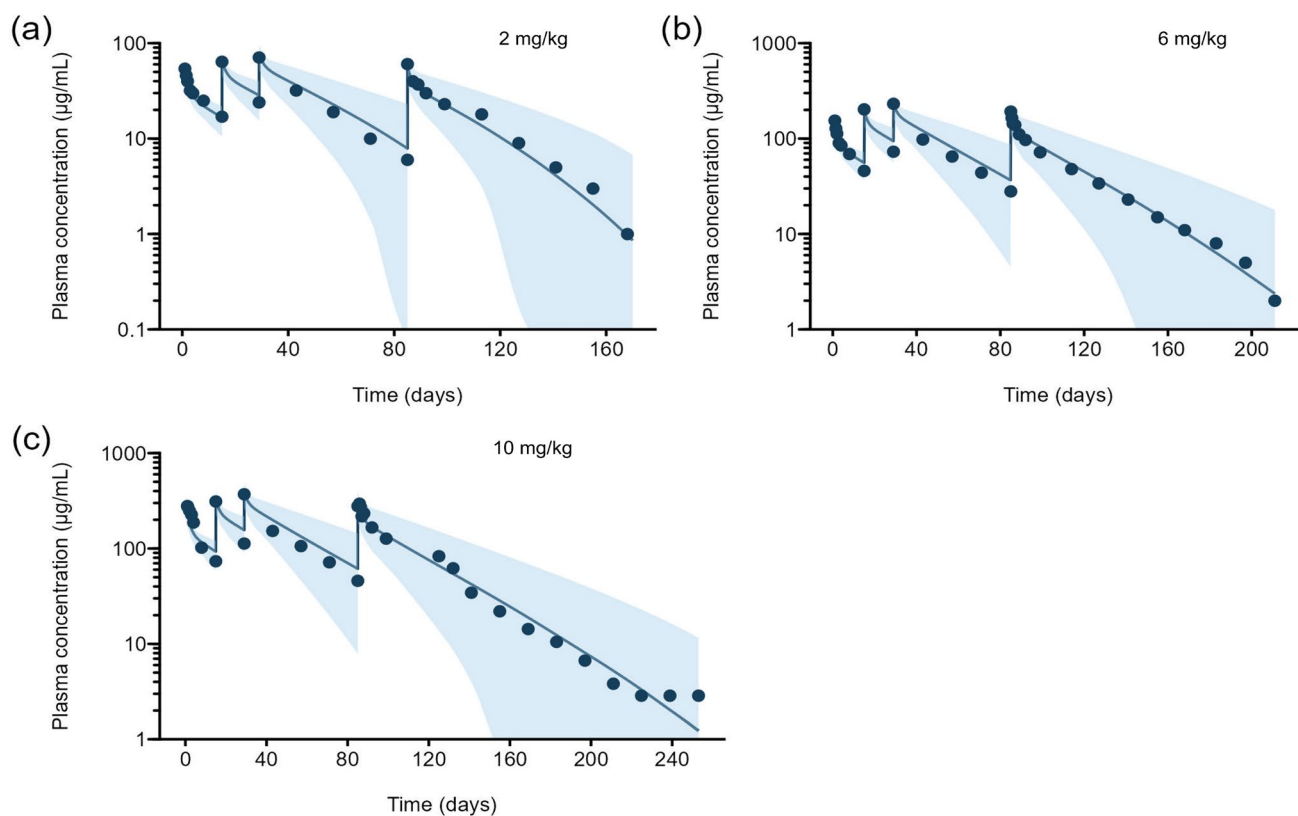


Fig. 3 Observed vs. predicted vedolizumab concentrations in patients with mild-to-moderate active ulcerative colitis. *Closed circles* represent the mean observed data³¹. *Solid lines* and *shaded areas* indicate the predicted mean and 95% prediction interval ($n = 100$ per dose group)

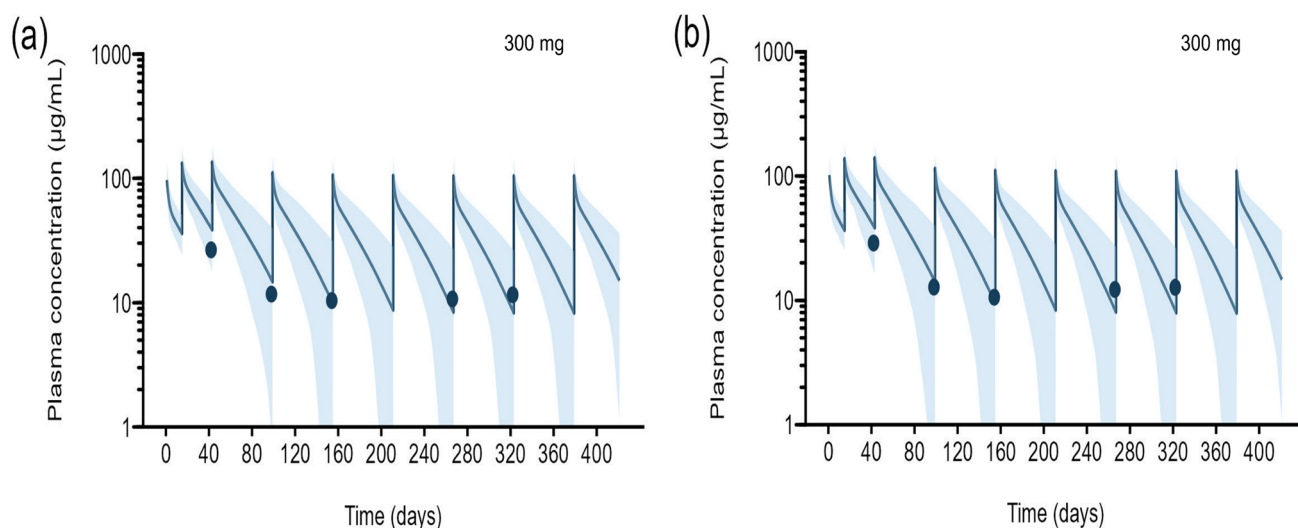


Fig. 4 Observed vs. predicted vedolizumab concentrations in patients with moderate-to-severe ulcerative colitis (**a**) and Crohn's disease (**b**). *Closed circles* represent the mean observed data²⁸. *Solid lines* and

shaded areas indicate the predicted mean and 95% prediction interval ($n = 100$ per group)

uncertain parameters have a significant impact on the model predictions. Finally, sensitive parameters were estimated using clinical data to further reduce uncertainty. Another

limitation of the current model is that the estimated $R_{\max}$ parameter may partially reflect multiple inflammation-related processes beyond $\alpha 4\beta 7$ $R_{\max}$ alone. Although

Table 3 Simulated and observed vedolizumab PK parameters following 2, 6, and 10 mg/kg multiple IV dosing in adults with mild-to-moderate active UC

Dose (mg/kg)		C_{max1d} (μg/mL)	AUC_{14d} (μg*day/mL)	C_{max85d} (μg/mL)	AUC_{85-99d} (μg*day/mL)	$AUC_{85-141d}$ (μg*day/mL)
2	Observed ^a	54	375	60.4	473	1082
	Predicted	47.7	319.3	55.2	426.8	842.9
	P/O ratio^b	0.88	0.85	0.91	0.90	0.78
6	Observed ^a	154.3	1058	191.9	1532	3318
	Predicted	141.4	995.7	176.3	1453.3	3249.3
	P/O ratio^b	0.92	0.94	0.92	0.95	0.98
10	Observed ^a	279	1765	291.9	2608	5633
	Predicted	263.3	1674.8	297	2459.1	5558.2
	P/O ratio^b	0.94	0.95	1.02	0.94	0.99

AUC_{14d} area under the curve from day 0 to 14, AUC_{85-99d} area under the curve from day 85 to 99, $AUC_{85-141d}$ area under the curve from day 85 to 141, C_{max1d} peak concentration at day 1, C_{max85d} peak concentration at day 85, *IV* intravenous, *PK* pharmacokinetic, *P/O* predicted-to-observed, *UC* ulcerative colitis

^aData from Parikh et al. [31]

^bP/O ratio calculated from the means of observed and simulated PK parameters

Table 4 Simulated and observed vedolizumab PK parameters following 300-mg IV dosing (induction and every 8 weeks maintenance) in adults with moderate-to-severe UC and CD

Type of IBD	Dose (mg)		C_{trough} Day 42 W=6 (μg/mL)	C_{trough} Day 98 W=14 (μg/mL)	C_{trough} Day 154 W=22 (μg/mL)	C_{trough} Day 266 W=38 (μg/mL)	C_{trough} Day 322 W=46 (μg/mL)
UC	300	Observed ^a	26.5	11.7	10.4	10.7	11.6
		Predicted	39.6	15.3	10.6	9.0	8.8
		P/O ratio^b	1.49	1.31	1.02	0.84	0.76
CD	300	Observed ^a	28.9	12.8	10.6	12.3	12.7
		Predicted	39.5	14.6	9.9	8.6	8.5
		P/O ratio^b	1.37	1.14	0.93	0.70	0.67

CD Crohn's disease, C_{trough} trough concentration at different time points, *CDER* Center for Drug Evaluation and Research, *FDA* Food and Drug Administration, *IBD* inflammatory bowel disease, *IV* intravenous, *PK* pharmacokinetic, *P/O* predicted-to-observed, *W* week, *UC* ulcerative colitis

^aData digitized from FDA CDER. Clinical Pharmacology Review Addendum, App. No. 125476orig1s000 (2013) [28]

^bP/O ratio calculated from the means of observed and simulated PK parameters

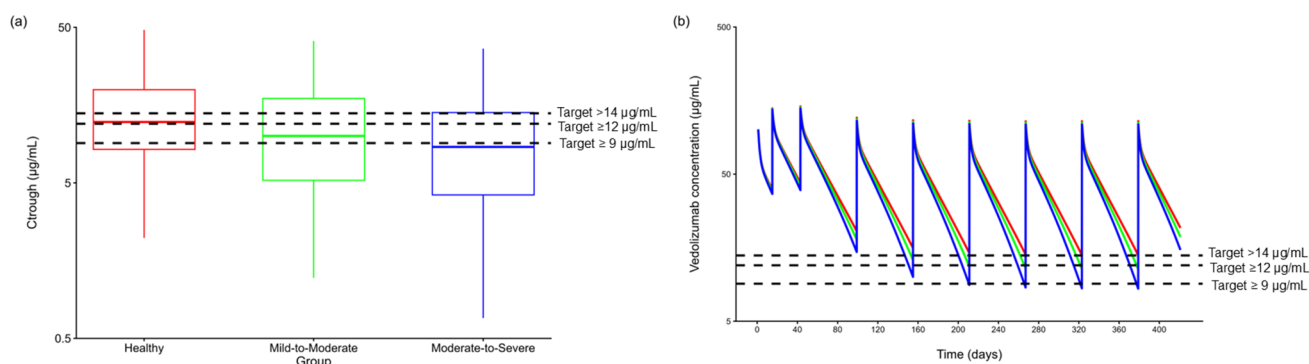


Fig. 5 Simulated vedolizumab steady-state C_{trough} and concentration-time profiles of each group following intravenous administration of 300 mg every 8 weeks ($n = 100$ per group). **a** Data are presented as median (*center line*), interquartile range (*box*), and 5th–95th percentiles (*whiskers*). The *dashed horizontal lines* represent the targets (9,

12, and 14 μg/mL). **b** The *red solid line* represents healthy volunteers; the *green solid line* represents patients with mild-to-moderate IBD; the *blue solid line* represents patients with moderate-to-severe IBD. C_{trough} trough concentration(s), *IBD* inflammatory bowel disease

modulation of $R_{\max}$ explained a significant portion of the observed PK variability between healthy adults and patients with IBD, other disease-related factors, including albumin levels, catabolic activity, vascular permeability, tissue distribution, and systemic clearance, were not explicitly incorporated into the present framework. In addition, these physiological factors may dynamically change over the course of treatment and remission, potentially influencing drug disposition over time. Future studies incorporating tissue-specific target expression and additional inflammation-driven physiological changes may further improve model identifiability and mechanistic interpretation. In addition, due to the lack of human organ-specific distribution data, an mPBPK framework was employed to characterize vedolizumab PK. This approach was selected to balance mechanistic relevance with practical model simplicity and parameter identifiability. Future studies incorporating tissue distribution data, particularly within target organs such as the gut, may enable more mechanistic characterization of site-of-action PK and further enhance the clinical relevance of the model.

5 Conclusion

In conclusion, this mPBPK model provides the first mechanistic framework for predicting vedolizumab PK across healthy and diseased populations. By incorporating the impact of target upregulation, the model explains the variability in exposure observed in clinical studies. This framework offers a valuable platform to facilitate dose individualization through MIPD and serves as a robust foundation for extrapolating PK to special populations.

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Declarations

Conflict of Interest The authors declare no conflicts of interest.

Data Availability All datasets used in this study are appropriately cited in the manuscript (see references).

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Code Availability The model simulation files are accessible upon reasonable request by contacting the corresponding author.

Author Contributions SP, KI, and TM wrote the manuscript, and all authors contributed to the revision of the draft manuscript. SP, KI, PM, RT, and TM designed the research, and SP, KI, TN, RT, and TM analyzed the data.

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